CLINICAL · EAR

Temporal bone fracture and CSF otorrhoea

A temporal bone fracture comes with a head injury, so trauma care comes first and the ear second. Once the patient is safe, three questions decide what happens next: is the face moving, can the patient hear, and is there CSF leaking?

AT A GLANCE

- Trauma care first.
- Record facial function and hearing early.
- Capsule-violating fractures do worst.
- Most CSF leaks stop; delayed palsy usually recovers.
- Classify as otic capsule-sparing or -violating; capsule-violating fractures risk late meningitis.

1 CHAPTER ONE Overview

A fracture through the hearing bone.

1.1 Epidemiology

Most fractures follow road traffic accidents, then assaults and falls. Young adult men in their second to fourth decades are affected about three times as often as women. A fracture is found in a substantial share of patients with significant head injury.

1.2 Why it matters

The temporal bone contains the cochlea, the labyrinth, the facial nerve, the ossicles, the carotid artery and the dura. A fracture can damage any of them, and early documentation of facial function and hearing shapes every later decision.

2 CHAPTER TWO Classification

Two systems, both still taught.

2.1 Traditional: by the direction of the fracture line

2.1.1 Longitudinal fracture

About 80%. Usually from a blow to the side of the head. The line runs along the long axis of the petrous bone, through the squamous part, the posterosuperior canal wall, the tegmen and the middle ear, and tends to spare the otic capsule.

2.1.2 Transverse fracture

About 20%. Usually from a blow to the back of the head. The line runs across the petrous bone, through the vestibule, cochlea or internal auditory canal.

2.1.3 Mixed

Elements of both, common in severe injuries.

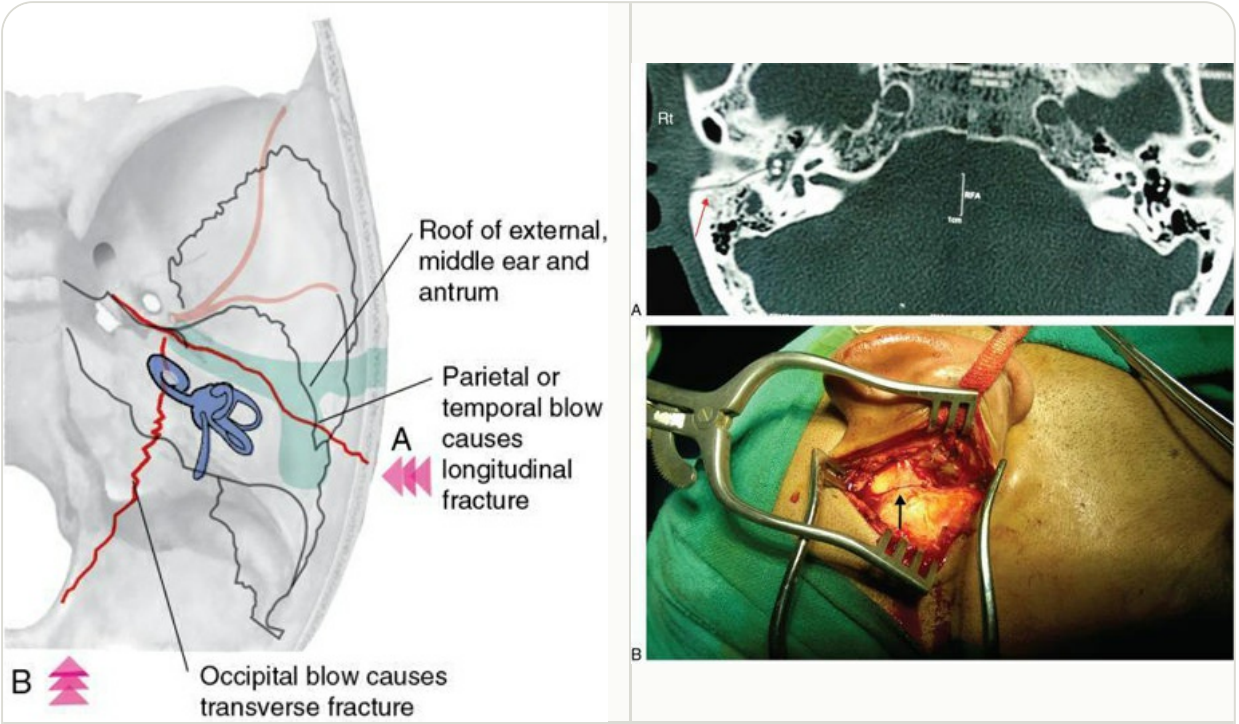


FIGURE 1 First, the courses of longitudinal and transverse fractures of the temporal bone. Second, a right longitudinal fracture on CT and as seen at operation.
Dhingra, figs 14.10 and 14.11

LONGITUDINAL AND TRANSVERSE FRACTURES

	LONGITUDINAL	TRANSVERSE
Frequency	About 80%	About 20%
Blow	Side of the head	Back of the head
Drum	Torn, bleeding from the ear	Intact, haemotympanum
Hearing	Conductive	Sensorineural
Vertigo	Less common	Severe
Facial palsy	About 20%, often delayed	About 50%, often immediate

2.2 Modern: does it involve the otic capsule?

Otic capsule sparing fractures are by far the commonest; *otic capsule violating* fractures, only about 6%, are much more likely to cause facial palsy, sensorineural hearing loss, CSF leak and meningitis. This system predicts outcome better.

WHERE THE BOOKS DIFFER · LONGITUDINAL OR TRANSVERSE

Dhingra
Teaches conductive loss with longitudinal and sensorineural loss with transverse fractures.

Lee
Modern series do not fit the old categories neatly; the otic capsule system predicts outcome better.

Know both
the old table is still examined.

3 CHAPTER THREE Clinical features

Look, listen, test the face.

3.1 General

Signs of head injury and of a base of skull fracture: *Battle's sign* (bruising over the mastoid, appearing after a day or two), periorbital bruising, and altered

consciousness.

3.2 The ear

Bleeding from the ear with a torn drum or a step in the canal wall (longitudinal), or a blue-black drum from blood in the middle ear (*haemotympanum*) behind an intact drum (transverse).

3.3 Hearing

3.3.1 Conductive loss

From haemotympanum (the commonest early cause), a perforation, or ossicular injury: incudostapedial joint separation first, then dislocation of the incus, then fracture of the stapes crura.

3.3.2 Sensorineural loss

From fracture through the cochlea or a labyrinthine concussion; often permanent. Mixed losses are common.

3.4 Vertigo

Severe vertigo with spontaneous nystagmus after a transverse fracture through the labyrinth; milder, positional vertigo after concussion or BPPV.

3.5 Facial nerve

Record whether the palsy was immediate or delayed, and complete or incomplete. An immediate complete palsy suggests transection or a bone spicule in the nerve; a delayed palsy suggests oedema and usually recovers.

3.6 CSF leak

Clear fluid from the ear (CSF otorrhoea) through a torn drum, or from the nose (CSF otorhinorrhoea) via the Eustachian tube behind an intact drum.

4 CHAPTER FOUR Investigations

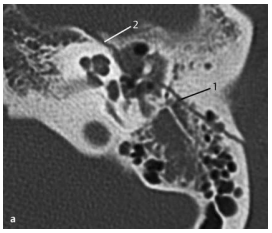
Image the bone and confirm the fluid.

4.1 High-resolution CT of the temporal bones

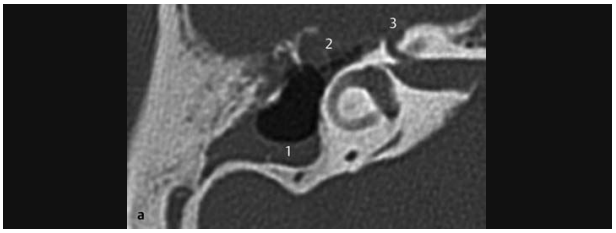
Shows the fracture line, the otic capsule, the ossicles, the facial canal and the tegmen. CT angiography if the fracture crosses the carotid canal.

IMAGING IN THE APP 2 cases

Skull base fractures, and dural prolapse with CSF leak.



Skull Base Fractures
Trauma · 13 images



Dural Prolapse and Cerebrospinal Fluid Leakage
After surgery · 2 images

4.2 Beta-2 transferrin

Confirms that the fluid is CSF.

4.3 Audiogram

As soon as practical, and repeated at six to eight weeks when blood has cleared from the middle ear.

4.4 Electroneuronography

For a complete facial palsy, to guide decisions on exploration.

5 CHAPTER FIVE Management

Trauma care first.

5.1 The first assessment

HEAD INJURY WITH EAR SIGNS

1. Follow trauma protocols and complete a neurological assessment.
2. Document facial function as early as possible, before sedation if you can.
3. Look for haemotympanum, a canal step, CSF, vertigo and nystagmus.
4. Do not syringe, probe or pack the ear deeply.
5. Arrange HRCT of the temporal bones, and CT angiography if the carotid canal is involved.

5.2 CSF leak

Most stop within a week or two with the head raised, bed rest, stool softeners and avoiding nose blowing and straining. Routine prophylactic antibiotics are not recommended. A persistent leak, or meningitis, needs a lumbar drain or surgical repair.

5.3 Facial palsy

Immediate complete palsy with evidence of nerve injury is explored and decompressed or repaired. Delayed palsy is managed with steroids and observation, with serial ENoG (see the facial nerve topic).

5.4 Hearing

Haemotympanum clears in weeks. A conductive loss persisting at six to eight weeks suggests ossicular injury and is explored and reconstructed. Sensorineural loss is rehabilitated with aids or, if profound, a cochlear implant.

OPERATIVE GUIDES IN THEATRE 4 operations

Temporal Bone Trauma
Temporal bone trauma
Cerebrospinal Fluid, Otorrhea and Encephalocele
CSF otorrhoea repair
Facial Nerve Decompression
Facial decompression
Ossicular Chain Reconstruction
Ossicular repair

5.5 Vertigo

Short-term vestibular sedatives, then vestibular rehabilitation; treat post-traumatic BPPV with repositioning manoeuvres.

6 CHAPTER SIX Complications

Early and late.

- **Meningitis,** early or years later, through an unhealed fracture.
- **Permanent sensorineural deafness.**
- **Facial palsy.**
- **Late cholesteatoma,** from skin trapped in the fracture line.
- **Chronic vertigo** and perilymph fistula.
- **Carotid injury** and dissection.

FIGURES AND LATE PROBLEMS (OXFORD TEXTBOOK)

The temporal bone is involved in 30 to 70% of head injuries, but only 14 to 22% of these produce a true fracture, mostly from road traffic accidents and falls. The old longitudinal–transverse scheme fails to describe the many mixed fractures, which is why *otic capsule-sparing* versus *otic capsule-violating* is now preferred: it predicts sensorineural loss, facial palsy and CSF leak better. Ossicular injury most often dislocates the incudostapedial joint (about 80%) and gives a conductive loss of at least 30 dB. About a quarter of fractures cause facial palsy (half of transverse and a tenth of longitudinal), and in about three-quarters of these the onset is delayed by one to sixteen days, which predicts a good recovery. A CSF leak complicates about 17%.

Bone of the otic capsule heals only by fibrous union, so capsule-violating fractures leave a weak barrier: CSF leaks can recur with coughing or straining, meningitis can occur years later, and meningoceles or encephaloceles may present long after the injury with a conductive loss, an effusion or meningitis. Patients with these fractures need pneumococcal vaccination and advice to seek help for fever and headache. Audiometry is best done two to three days after injury (after the patient wakes) and repeated; the Sunnybrook scale records synkinesis better than House–Brackmann during recovery.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Longitudinal versus transverse:** the six-row table.
- **Otic capsule sparing versus violating,** and why it predicts outcome.
- **Immediate versus delayed facial palsy,** and how each is managed.
- **Beta-2 transferrin** for CSF.
- **Ossicular injuries** in order of frequency.

7.2 Common pitfalls

COMMON PITFALLS

- Forgetting to record facial function before the patient is sedated.
- Syringing or packing a bleeding ear after head injury.
- Assuming a conductive loss is permanent before the blood has cleared.
- Missing late meningitis or cholesteatoma years after the injury.

Written for this app from Oxford Textbook of Otolaryngology ch. 23, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 25, Dhingra ch. 14 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

TEMPORAL BONE & SKULL BASE · TRAUMA

Skull Base Fractures

IMAGING FINDINGS

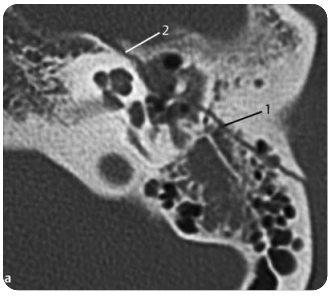
- Fracture line through the temporal bone or skull base; air in the middle ear or cranium
- Otic capsule sparing (most) or violating
- Fluid in the middle ear and mastoid

TOP DIFFERENTIALS

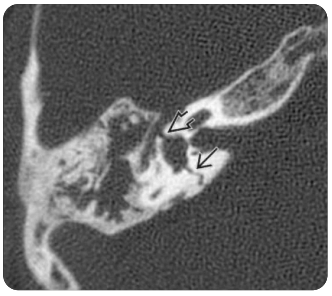
- Sutures, fissures and small canals (pseudofractures)
- Vascular grooves

CHECKLIST

- Ossicles, facial canal (geniculate), tegmen and CSF leak
- Carotid canal involved: CT angiography



CT, axial. Patient with head trauma showing a fracture line through the mastoid (1) extending just anterior to the cochlea (2). Luxation of the…



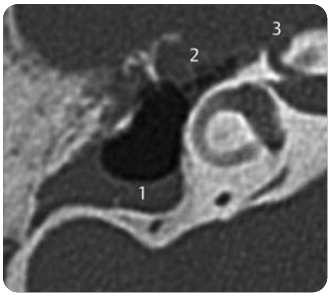
Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

MIDDLE EAR & MASTOID · AFTER SURGERY

Dural Prolapse and Cerebrospinal Fluid Leakage

- Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.



Cerebrospinal fluid leakage after insertion of a tympanostomy tube (grommet), indicated for middle ear secretions and conductive hearing…